BYoung Physical Therapy

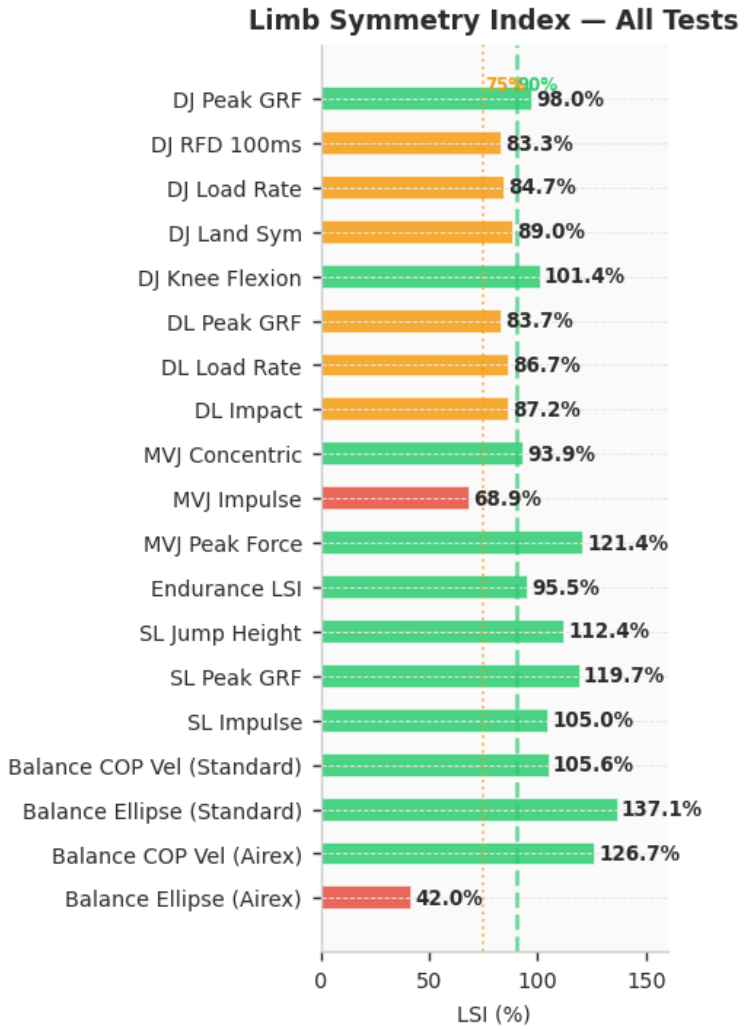
Return to Sport Assessment

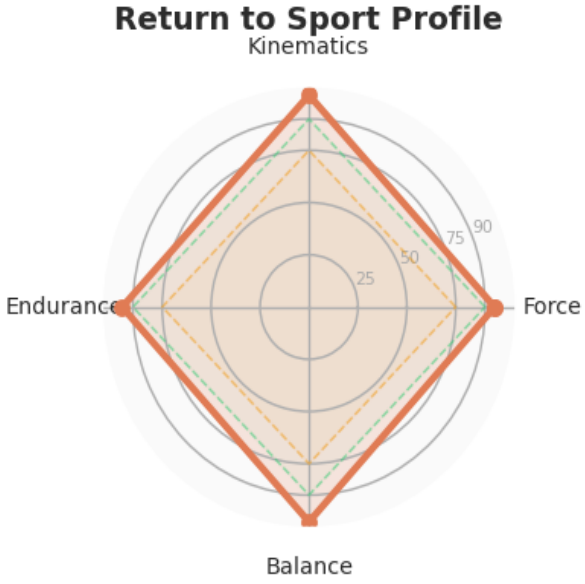
Patient: Alex Walker **Surgical Side:** L **Date:** April 02, 2026 **Surgery:** 08/25 (8.1 months post-op)

55

RTR Score
Caution
8 mo post-op

Metric	Surgical	Non-Surg
DJ Peak GRF	856.9	874.8
DJ RFD 100ms	7759.8	9317.4
DJ Load Rate	9568.7	11294.7
DJ Land Sym	—	—
DJ Knee Flexion	97.1	95.7
DL Peak GRF	660.4	789.3





CLINICAL FLAGS

■ **Time post-op:** 8.1 months post-op — below the 9-month evidence threshold. Risk is 4–7× higher than athletes who wait ≥ 9 months (Kyritsis 2016 BJSM). Grade capped at Caution. [Time readiness: 69%]

■ Score: Performance 80/100 → Time modifier ×0.69 → Final 55/100

• MVJ Impulse: LSI 68.9% — significant asymmetry

• Balance Ellipse (Airex): LSI 42.0% — significant asymmetry

● ≥90% Symmetric ● ≥75% Progressing ● <75% Deficit LSI = Surgical ÷ Non-Surgical × 100

DROP JUMP

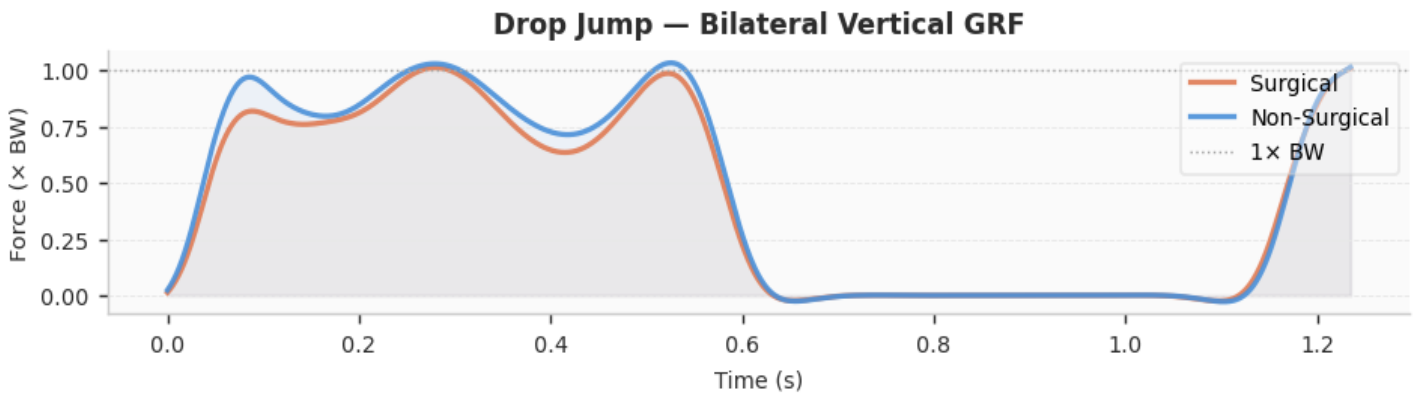
Metric	Surgical	Non-Surg	LSI
Peak GRF (N)	856.9	874.8	98.0%
Impulse (N·s)	438.6	468.3	93.7%
RFD 100ms (N/s)	7759.8	9317.4	83.3%
Loading Rate 50ms (N/s)	9568.7	11294.7	84.7%
Landing Sym 200ms (%)	—	—	89.0%
Impact Transient (N)	489.0	584.7	83.6%
Time-to-Peak (ms)	279.0	526.0	—
RSI	1.540	—	—
Contact Time (s)	0.623	—	—
Peak Knee Flexion (°)	97.1	95.7	101.4%
Peak Knee Valgus (°)	1.7	4.2	41.0%

CLINICAL INTERPRETATION

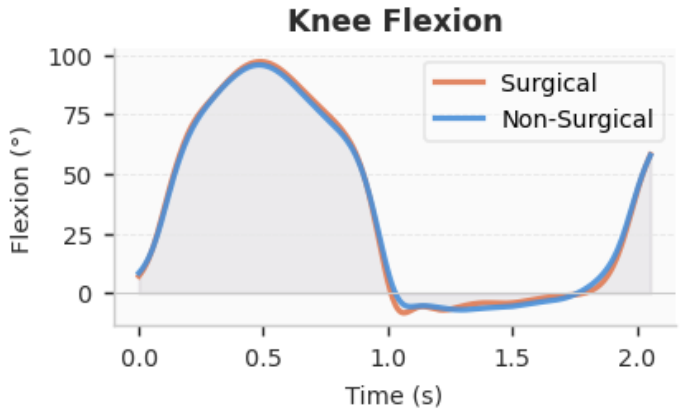
RSI 1.54 (✓ meets RTR target ≥ 1.30). Reactive capacity is adequate.

Landing symmetry 89% LSI: Surgical limb loading 11% less than non-surgical — surgical guarding detected.

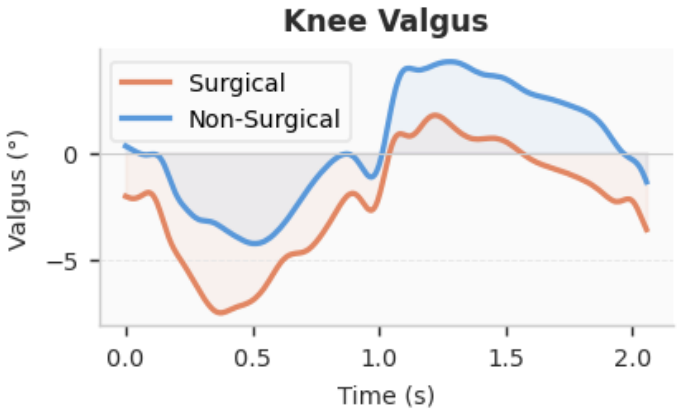
Targets: RSI ≥ 1.30 · Landing LSI 90–110% · Load rate LSI ≥ 90% · Knee flexion ≥ 60° · Valgus < 5°.



Bilateral GRF (× body weight). Orange = surgical, Blue = non-surgical.



Knee flexion. Target ≥ 60° at contact.



Knee valgus. Target < 5°. Elevated valgus = ACL re-injury risk.

DROP LANDING

Metric	Surgical	Non-Surg	LSI
Peak GRF (N)	660.4	789.3	83.7%
Impulse (N·s)	447.5	484.0	92.5%
Loading Rate 50ms (N/s)	10087.3	11640.7	86.7%
Impact Transient (N)	526.3	603.5	87.2%
Peak Knee Flexion (°)	102.5	103.8	98.8%
Peak Knee Valgus (°)	-0.5	1.2	-37.3%

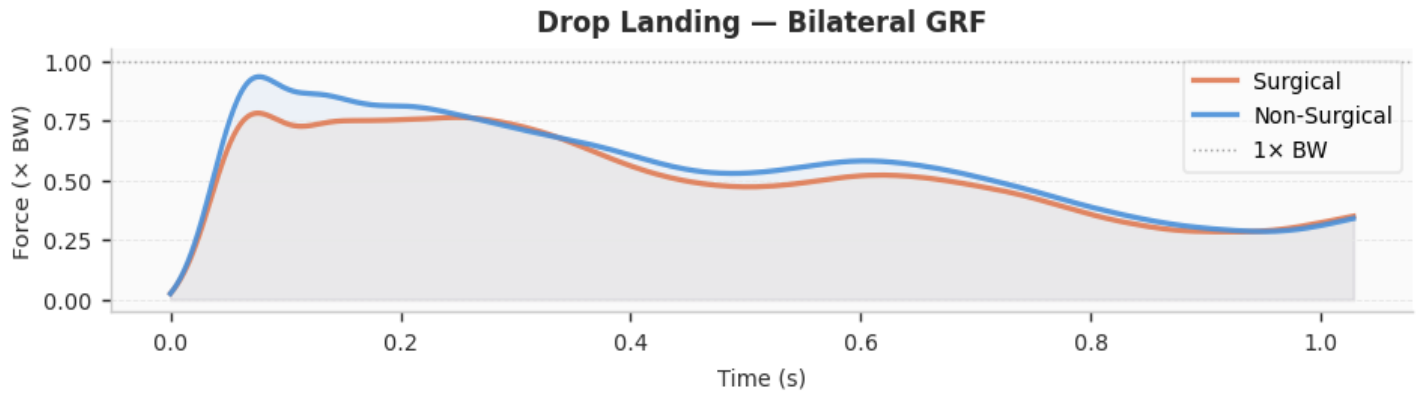
CLINICAL INTERPRETATION

Absorptive landing — assesses shock attenuation without a re-jump. Lower, more gradual force = better control.

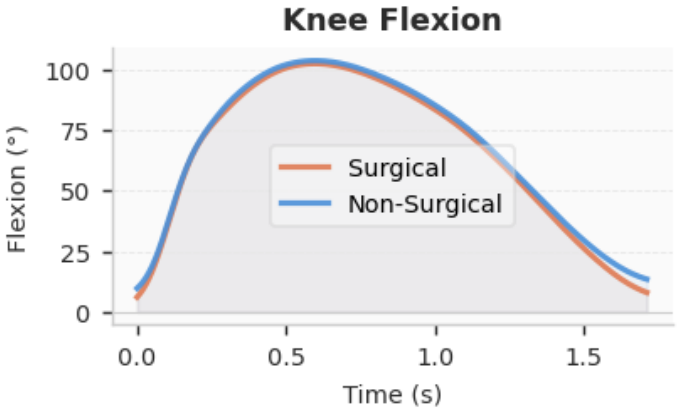
Load rate LSI 87%: Surgical limb loading rate 13% lower — surgical inhibition.

Impact transient LSI 87%. Target: < 2× BW peak, LSI 90–110%, no sharp spike.

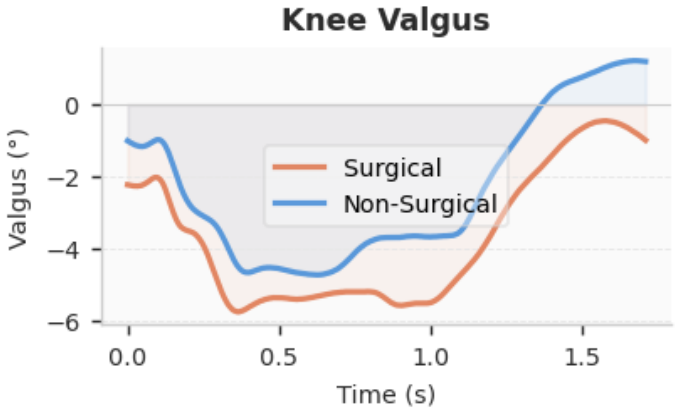
Targets: Peak GRF < 3× BW · Load rate LSI 90–110% · Knee flexion ≥ 60° · Valgus < 5°.



Bilateral GRF. Smooth progressive rise = good shock absorption. Sharp spike = stiff landing.



Knee flexion at landing.



Knee valgus at landing. > 8° surgical = elevated ACL stress.

MAX VERTICAL JUMP (CMJ)

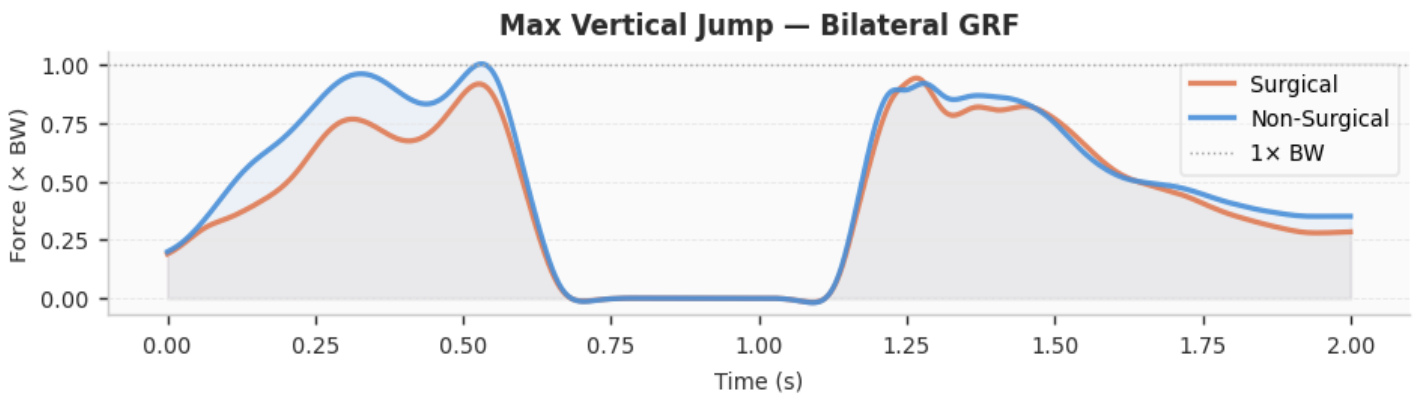
Metric	Surgical	Non-Surg	LSI
Peak Concentric GRF (N)	796.1	847.6	93.9%
Propulsion Impulse (N·s)	345.2	500.8	68.9%
Peak Force Surg / NS (N)	629.4	518.6	121.4%
Propulsion LSI (%)	—	—	68.9%
Jump Height	18.4	—	—
Flight Time (s)	0.442	—	—
Unweighting Imp Surg/NS	290.6	262.5	—
Peak Knee Flexion (°)	103.7	104.0	99.7%
Peak Knee Valgus (°)	1.3	4.4	30.9%

CLINICAL INTERPRETATION

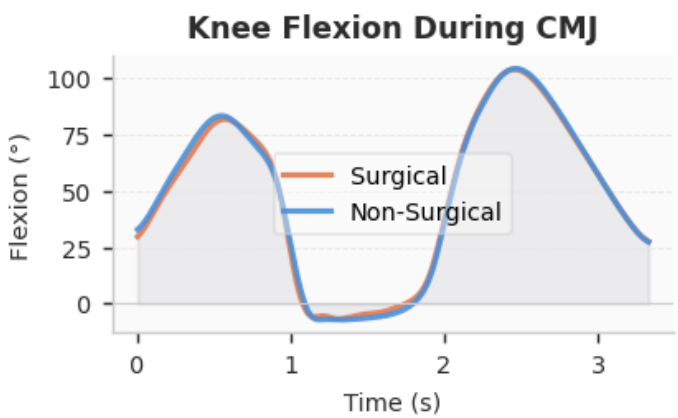
Bilateral CMJ — sensitive to subtle propulsion deficits masked at lower intensities.

Peak force LSI 121%: Surgical limb generating 21% more force — possible NS compensation.

Propulsion LSI 69%. Targets: ≥ 90% propulsion, ≥ 90% peak force, matched unweighting impulse.



Full CMJ cycle: unweighting → eccentric → propulsion → flight. Propulsive peaks should be symmetric.



Knee flexion depth during CMJ. Asymmetric depth = compensatory strategy.

ENDURANCE SQUAT (30s)

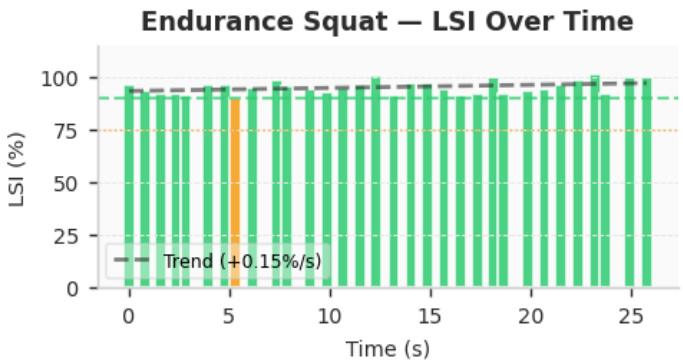
Metric	Surgical	Non-Surg	LSI
Mean Peak Force LSI (%)	95.5	—	95.5%
LSI First 10s (%)	94.4	—	94.4%
LSI Last 10s (%)	97.1	—	97.1%
Fatigue Drift (Δ%)	2.7	—	—
Total Reps Detected	32.0	—	—
Mean Peak Force Surg (N)	831.8	—	—
Mean Peak Force NS (N)	872.4	—	—

CLINICAL INTERPRETATION

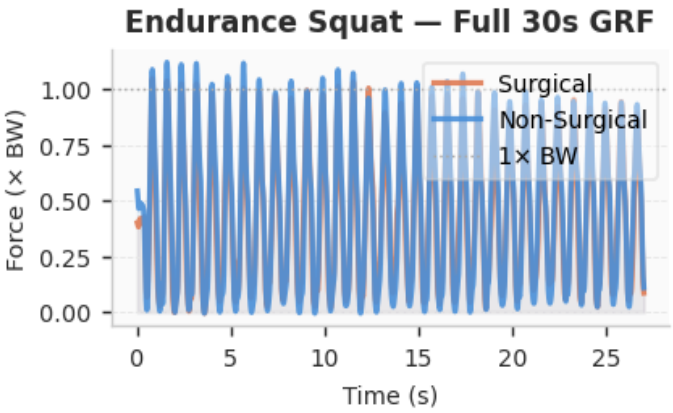
30s bilateral squat — reveals neuromuscular fatigue-induced asymmetry missed by single-rep testing.

Fatigue drift 2.7%: ✓ Symmetry maintained under fatigue.

Targets: Mean LSI ≥ 90% · Fatigue drift < 5% · No progressive offloading across 30s.



Per-rep LSI across 30s. Declining trend = neuromuscular fatigue breakdown.



Full 30s GRF trace. Force magnitude and rhythm across the set.

SINGLE-LEG VERTICAL JUMP

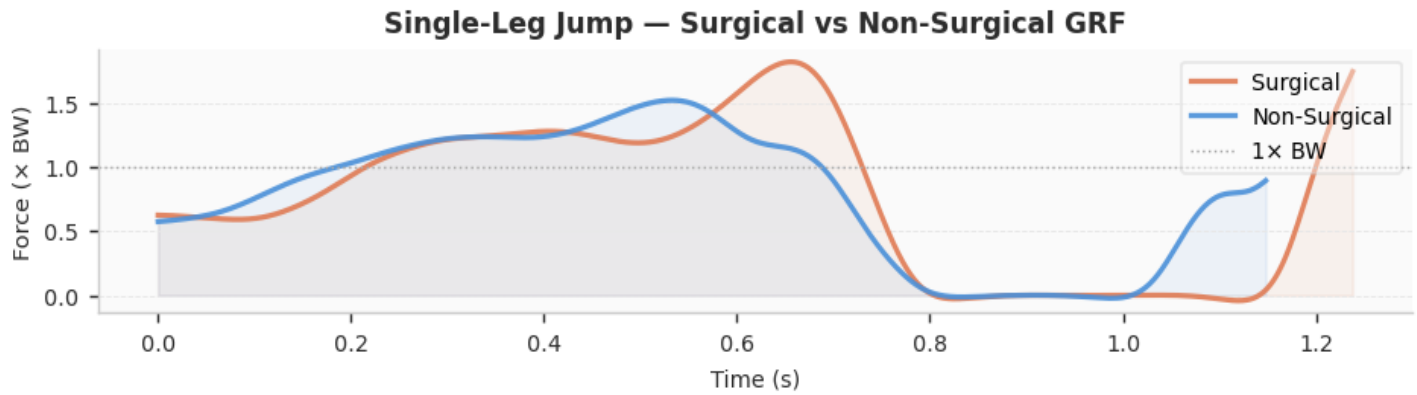
Metric	Surgical	Non-Surg	LSI
Jump Height	11.2	9.9	112.4%
Peak GRF (N)	1536.7	1284.1	119.7%
Propulsion Impulse (N·s)	730.5	695.6	105.0%
RFD 100ms (N/s)	—	1505.4	—
Flight Time (s)	0.3	0.2	—
Contact Time (s)	0.8	0.8	—
Peak Knee Flexion (°)	59.2	50.6	—
Peak Knee Valgus (°)	-0.7	2.2	—

CLINICAL INTERPRETATION

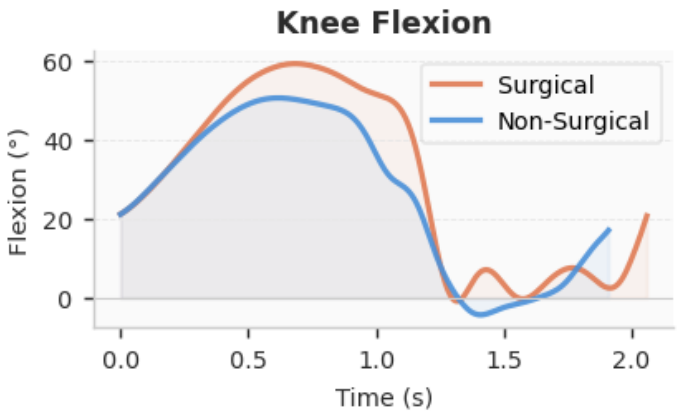
Highest-demand test — isolates each limb, removing bilateral compensation. Strongest re-injury risk predictor.

Jump height LSI 112%: ✓ Meets RTR threshold (≥ 90%).

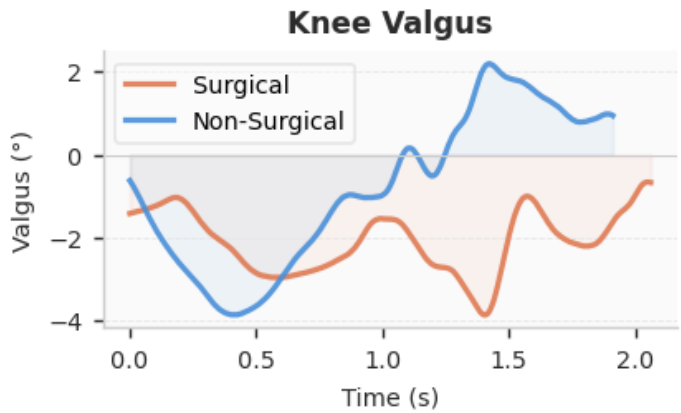
Targets: Jump height LSI ≥ 90% · Peak GRF LSI ≥ 90% · Valgus < 5° on both limbs.



Overlaid GRF for each limb tested separately. Propulsive peak height and shape should match.



Knee flexion strategy per limb.



Knee valgus per limb. Elevated surgical valgus under single-leg load = re-injury risk.

BALANCE — STANDARD (EYES CLOSED, FIRM SURFACE)

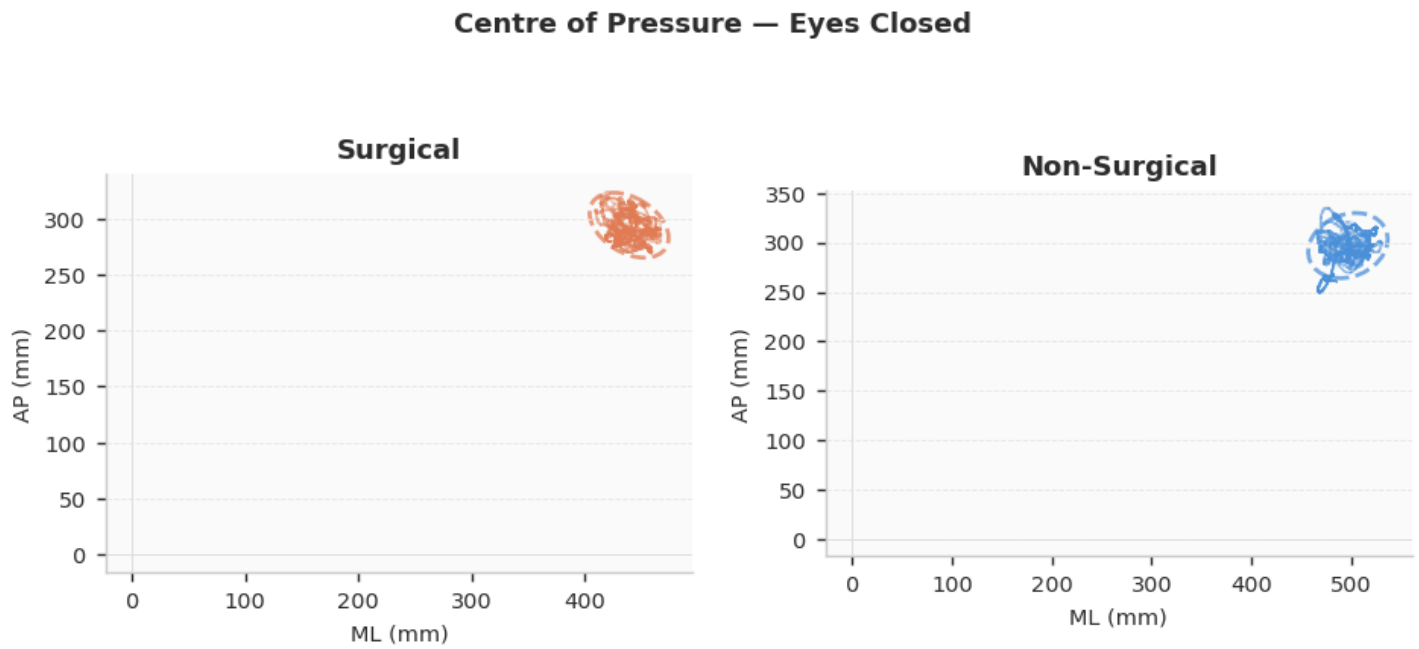
Metric	Surgical	Non-Surg	LSI
COP Mean Velocity (mm/s)	430.6	454.5	105.6%
COP Ellipse Area (mm²)	2943.3	4036.6	137.1%
Total Excursion (mm)	6027.5	6362.9	—
AP Range (mm)	54.3	85.9	—
ML Range (mm)	58.3	65.4	—
RMS Displacement (mm)	18.6	21.1	—

CLINICAL INTERPRETATION

Eyes-closed single-leg stance on firm surface. COP velocity is the most sensitive metric — higher speed = more corrective activity = reduced proprioception. Target: < 12 mm/s.

Surgical 431 mm/s vs Non-Surgical 455 mm/s — COP velocity comparable between limbs.

COP Velocity LSI 106% (NS÷Surg×100 — lower = better). Normative target: < 12 mm/s. LSI target ≥ 90%.



COP trajectory — eyes closed. Tighter ellipse = better joint position sense. 95% confidence ellipse shown.

BALANCE — AIREX PAD (EYES CLOSED, UNSTABLE SURFACE)

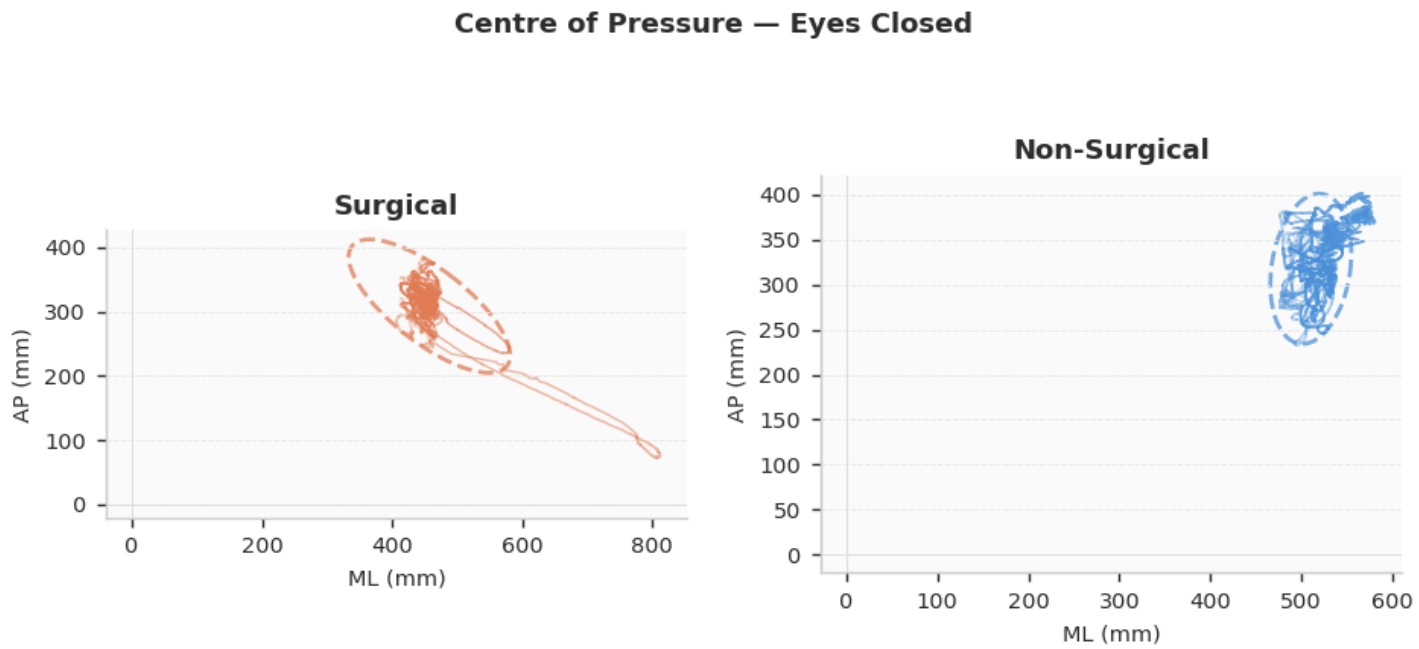
Metric	Surgical	Non-Surg	LSI
COP Mean Velocity (mm/s)	573.7	726.9	126.7%
COP Ellipse Area (mm²)	27205.6	11439.1	42.0%
Total Excursion (mm)	8031.6	10163.4	—
AP Range (mm)	311.1	168.7	—
ML Range (mm)	406.1	105.6	—
RMS Displacement (mm)	66.1	38.7	—

CLINICAL INTERPRETATION

Eyes-closed single-leg stance on Airex foam pad. Higher COP values are expected — the unstable surface challenges proprioceptive and neuromuscular systems. Symmetry between limbs (LSI ≥ 85%) is the key target.

Surgical 574 mm/s vs Non-Surgical 727 mm/s — COP velocity comparable between limbs.

COP Velocity LSI 127% (NS÷Surg×100 — lower = better). Normative target: < 50 mm/s. LSI target ≥ 90%.



COP trajectory — eyes closed. Tighter ellipse = better joint position sense. 95% confidence ellipse shown.